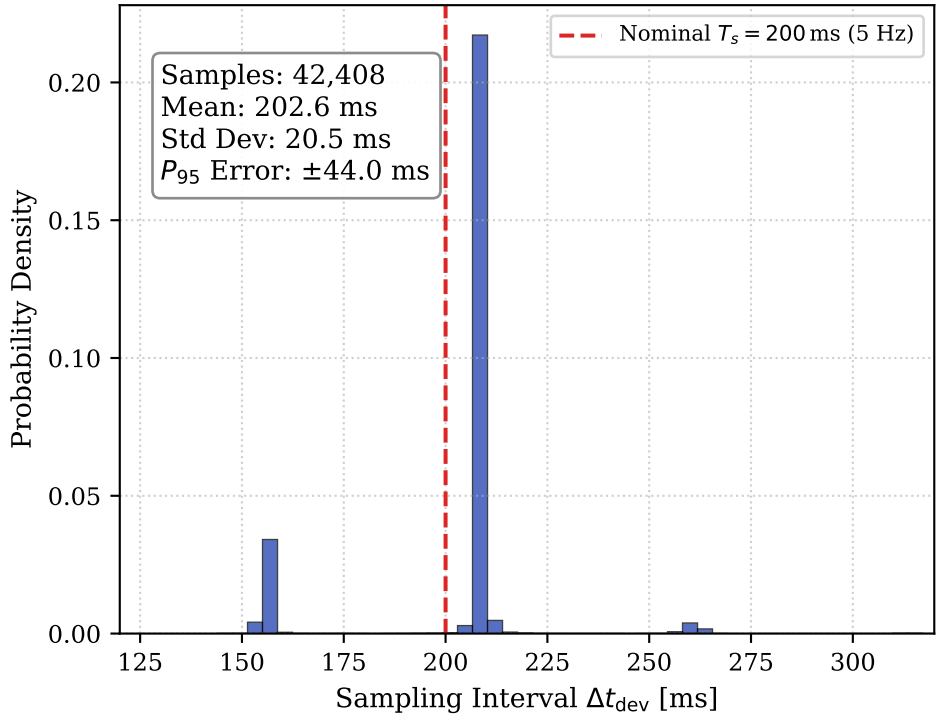
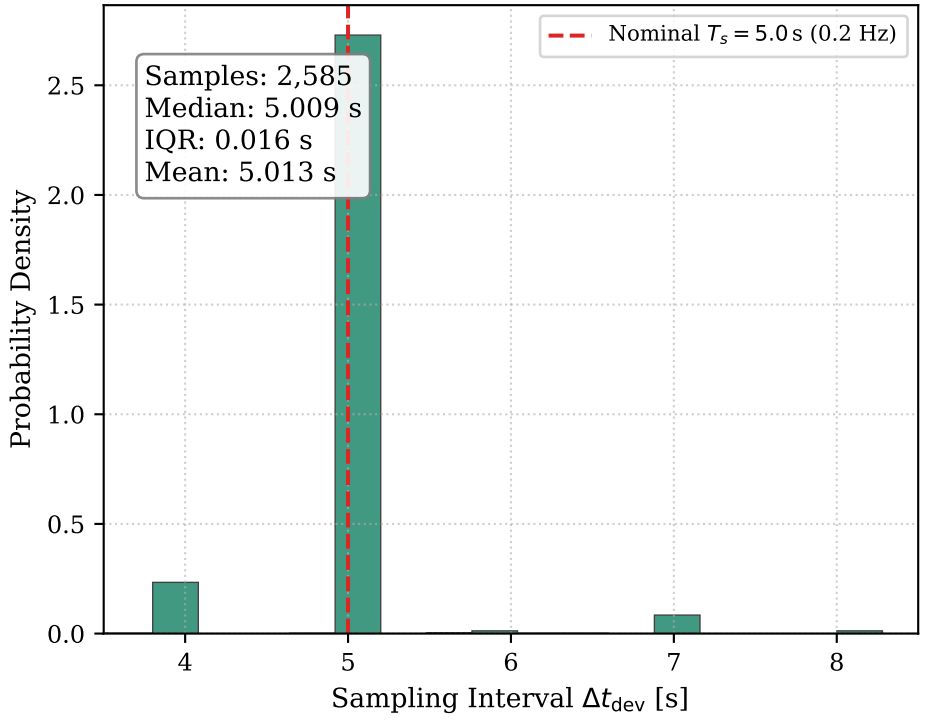

(a) Aerator Node: Actuation Interval (Δt_{dev})
Nominal $T_s = 200$ ms (5 Hz)
Samples: 42,408
Mean: 202.6 ms
Std Dev: 20.5 ms
P_{95} Error: ±44.0 ms
Probability Density
0.00
0.05
0.10
0.15
0.20
125
150
175
200
225
250
275
300
Sampling Interval Δt_{dev} [ms]


(b) Optical DO Sensor: Modbus Polling (Δt_{dev})
Nominal $T_s = 5.0$ s (0.2 Hz)
Samples: 2,585
Median: 5.009 s
IQR: 0.016 s
Mean: 5.013 s
Probability Density
0.0
0.5
1.0
1.5
2.0
2.5
4
5
6
7
8
Sampling Interval Δt_{dev} [s]